Dielétricos:

Comprimento de onda de corte: $\lambda_c = 2d\,NA = 2d\sin\bar{\theta}_c$

Frequência de corte: $\nu_c = \frac{1}{NA}\frac{c_0}{2d}$

Frequência angular de corte: $\omega_c = \frac{1}{NA}\frac{\pi c_0}{d}$

Nº de modos: $M \doteq \frac{\sin\bar{\theta}_c}{\frac{\lambda}{2d}} \Rightarrow M \doteq \frac{2d}{\lambda_0} NA$

Monomodal: $M = 1$ $\quad \frac{\sin\bar{\theta}_c}{\frac{\lambda}{2d}} < 1$ (Há sempre modos guiados)

Condição de auto-consistência - TE: $tg\left(\frac{\pi d}{\lambda}\sin\theta - m\frac{\pi}{2}\right) = \sqrt{\frac{\sin^2\bar{\theta}_c}{\sin^2\theta} - 1}$

Condição de auto-consistência - TM: $tg\left(\frac{\pi d}{\lambda}\sin\theta - m\frac{\pi}{2}\right) = -\frac{1}{\cos^2\theta_c}\sqrt{\frac{\sin^2\bar{\theta}_c}{\sin^2\theta} - 1}$

$\sin\theta_c = \frac{n_2}{n_1} \Leftrightarrow \cos\bar{\theta}_c = \frac{n_2}{n_1}$ $\qquad \bar{\theta}_c = 90° - \theta_c$

$\sin\theta_a = NA = \sqrt{n_1^2 + n_2^2}$

1. a) $n_1 \sin\theta_1 = n_2 \sin\overset{\to 90°}{\theta_2}$

$\Rightarrow \sin\theta_c = \frac{n_2}{n_1} \Leftrightarrow \theta_c = 61{,}05°$

$\bar{\theta}_c = \frac{\pi}{2} - \theta_c = 28{,}96°$

$NA = \sqrt{n_1^2 + n_2^2} = 0{,}77$

$\sin\theta_a = NA \Leftrightarrow \theta_a = 50{,}77°$

b) $M \doteq \frac{\sin\bar{\theta}_c}{\frac{\lambda}{2d}} = \frac{\sin\bar{\theta}_c}{\frac{\lambda_0}{2n_1 d}} = 3{,}56 \doteq 4$

$tg\left(\frac{\pi d}{\lambda}\sin\theta - m\frac{\pi}{2}\right) = \sqrt{\frac{\sin^2\bar{\theta}_c}{\sin^2\theta} - 1}$

$g(x) = \sqrt{\frac{\sin^2\bar{\theta}_c}{\sin^2\theta} - 1} = \sqrt{\frac{\sin^2\bar{\theta}_c}{x^2} - 1}$

$x \to \sin\bar{\theta}_c \Rightarrow f(x) \to 0$

$x \to 0 \Rightarrow f(x) \to +\infty$

$$f(x) = tg\left(\frac{\pi d}{\lambda}x - m\frac{\pi}{2}\right) = \begin{cases} tg\left(\frac{\pi d}{\lambda}x\right) & , m \text{ par} \to m = 2n \\ -\cot\left(\frac{\pi d}{\lambda}x\right) & , m \text{ ímpar} \to m = 2n+1 \end{cases}$$

m Par:

zeros: $tg\left(\frac{\pi d}{\lambda}x\right) = 0 \Rightarrow \frac{\pi d}{\lambda}x = n\pi \Leftrightarrow x = \frac{n d}{\lambda} \Leftrightarrow x = \frac{2nd}{2\lambda} \Rightarrow x = \frac{md}{2\lambda}$

extremos: $tg\left(\frac{\pi d}{\lambda}x\right) = +\infty \Rightarrow \frac{\pi d}{\lambda}x = \frac{\pi}{2} + n\pi \Leftrightarrow x = \left(\frac{1}{2} + n\right)\frac{\lambda}{d} \Leftrightarrow (1+2n)\frac{\lambda}{2d} = x \Leftrightarrow x = (m+1)\frac{\lambda}{2d}$

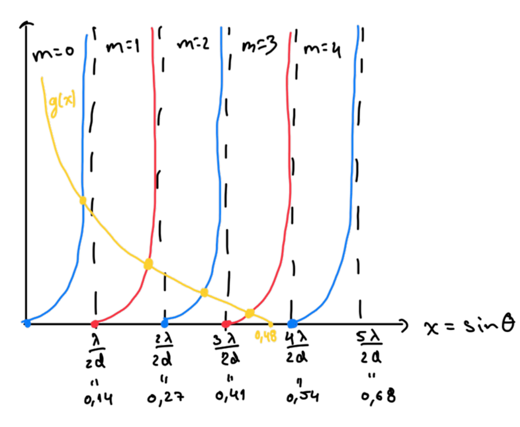
$m = 0, 2, 4, 6 \ldots$

m Ímpar:

zeros: $-\cot\left(\frac{\pi d}{\lambda}x\right) = 0 \Rightarrow \frac{\pi d}{\lambda}x = \frac{\pi}{2} + n\pi \Leftrightarrow x = \left(\frac{1}{2} + n\right)\frac{\lambda}{d} \Leftrightarrow x = (2n+1)\frac{\lambda}{2d} \Leftrightarrow x = m\frac{\lambda}{2d}$

extremos: $-\cot\left(\frac{\pi d}{\lambda}x\right) = +\infty \Rightarrow \cot\left(\frac{\pi d}{\lambda}x\right) = -\infty \Leftrightarrow \frac{\pi d}{\lambda}x = \pi + n\pi \Leftrightarrow x = (1+n)\frac{\lambda}{d} \Leftrightarrow x = (2n+2)\frac{\lambda}{2d} \Leftrightarrow x = (2n+1+1)\frac{\lambda}{2d} \Leftrightarrow x = (m+1)\frac{\lambda}{2d}$

$m = 1, 3, 5, 7, \ldots$




$\sin\theta = \frac{m\lambda}{2d}$ $\quad m = 0 \Rightarrow \sin\theta = 0$ $\quad m = 1 \Rightarrow \sin\theta = \frac{\lambda}{2d}$ $\quad m_2 = 2 \Rightarrow \sin\theta = \frac{2\lambda}{2d}$ $\quad m = 3 \Rightarrow \sin\theta = \frac{3\lambda}{2d}$ $\quad m = 4 \Rightarrow \sin = 4\frac{\lambda}{2d}$

tan (m par $_{0,2,4}$) zeros $= \frac{m\lambda}{2d} \Rightarrow 0, \frac{2\lambda}{2d}, \frac{4\lambda}{2d}$

cot (m ímpar $_{1,3}$) zeros: $\frac{\lambda}{2d}, \frac{3\lambda}{2d}$

tan (m par $_{0,2,4}$) extremos: $\frac{\lambda}{2d}, \frac{3\lambda}{2d}, \frac{5\lambda}{2d}$

cot (m ímpar $_{1,3}$) extremo: $\frac{2\lambda}{2d}, \frac{4\lambda}{2d}$

g(x) zero: $x = \sin\bar{\theta}_c = 0{,}48$

2. $\sin\theta_c = \frac{n_2}{n_1} = 0{,}625 \to \theta_c = 38{,}68°$

$\bar{\theta}_c = 51{,}32$ $\qquad \sin\bar{\theta}_c = 0{,}78$

$M \doteq \frac{2d\sin\bar{\theta}_c}{\lambda} \doteq \frac{2d\,n_1\sin\bar{\theta}_c}{\lambda_0} = 5{,}74 \doteq 6$

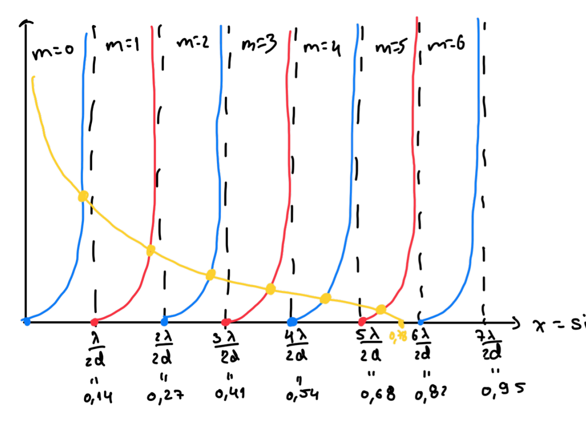
tan (m par $_{0,2,4,6}$) zeros $= \frac{m\lambda}{2d} \Rightarrow 0, \frac{2\lambda}{2d}, \frac{4\lambda}{2d}, \frac{6\lambda}{2d}$

cot (m ímpar $_{1,3,5}$) zeros: $\frac{\lambda}{2d}, \frac{3\lambda}{2d}, \frac{5\lambda}{2d}$

tan (m par $_{0,2,4,6}$) extremos: $\frac{\lambda}{2d}, \frac{3\lambda}{2d}, \frac{5\lambda}{2d}, \frac{7\lambda}{2d}$

cot (m ímpar $_{1,3,5}$) extremo: $\frac{2\lambda}{2d}, \frac{4\lambda}{2d}, \frac{6\lambda}{2d}$

g(x) zero: $x = \sin\bar{\theta}_c = 0{,}78$




3.

| |
|---|
| $n_2 = 1{,}4$ |
| $n_1 = 1{,}6$ |
| $n_3 = 1{,}3$ |

2-1:

$\cos\bar{\theta}_{c2} = \frac{n_2}{n_1} = 0{,}875 \Rightarrow \bar{\theta}_{c2} = 28{,}96°$

$NA = \sqrt{1{,}6^2 - 1{,}4^2} = 0{,}77$

$\theta_a = \arcsin(NA) = 50{,}35°$

$M \doteq \frac{\sin\bar{\theta}_c}{\frac{\lambda}{2d}} = \frac{0{,}48 \times 1{,}6 \times 2 \times 2}{0{,}87} = 3{,}53 \doteq 4$

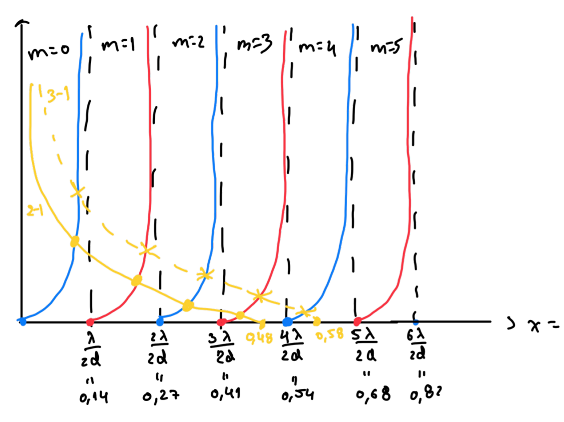
3-1:

$\cos\bar{\theta}_{c3} = \frac{n_3}{n_1} = 0{,}8125 \Rightarrow \bar{\theta}_{c3} = 35{,}66°$

$NA = \sqrt{1{,}6^2 - 1{,}3^2} = 0{,}93$

$\theta_a = \arcsin(NA) = 68{,}87°$

$M \doteq \frac{0{,}58 \times 1{,}6 \times 2 \times 2}{0{,}87} = 4{,}29 \doteq 5$




2-1: g(x) zero $= 0{,}48$ $\qquad x = \sin^2\bar{\theta}_c$

3-1: g(x) zero $= 0{,}58$

4. $M = 1$ $\quad M = \frac{\sin\bar{\theta}_c\, 2d}{\lambda} \Leftrightarrow 1 > \frac{\sin\bar{\theta}_c\, 2d n_1}{\lambda_0} \Leftrightarrow d > \frac{\lambda_0}{2n_1\sin\bar{\theta}_c} \Rightarrow d = 1{,}88\ \mu m$

$\cos\bar{\theta}_c = \frac{n_2}{n_1} = 0{,}973$ $\qquad \sin\bar{\theta}_c = 0{,}23$

$\bar{\theta}_c = 13{,}26$

$M \doteq \frac{\sin\bar{\theta}_c\, 2d n_1}{\lambda_0} = 1{,}53 \doteq 2$ modos

5. $\lambda_c = \frac{2d}{m} NA \Leftrightarrow \lambda_c = \frac{2d}{m}\sqrt{n_1^2 - n_2^2} \Leftrightarrow \lambda_c^2 = \frac{4d^2}{m^2}(n_1^2 - n_2^2) \Leftrightarrow \lambda_c^2 = \frac{4d^2}{m^2}\underbrace{(n_1 - n_2)}_{\Delta n}(n_1 + n_2) \Leftrightarrow \lambda_0^2 = \frac{4d^2}{m^2}\Delta n\, 2n_1 \Leftrightarrow \lambda_0^2 = 8 n_1 \Delta n \frac{d^2}{m^2}$ $\quad n_1 \approx n_2$

6. $\frac{2\pi}{\lambda}(2d\sin\theta) - 2\varphi_r = 2\pi m$

$E = ZH = \frac{Z_0}{n}H$

$\begin{cases} E_{i,\parallel} = -E_i\cos\theta_i \\ E_{r,\parallel} = E_r\cos\theta_r \\ E_{t,\parallel} = -E_t\cos\theta_t \end{cases}$ $\quad \begin{cases} H_i + H_r = H_t \\ E_{i,\parallel} + E_{r,\parallel} = E_{t,\parallel} \end{cases} \Leftrightarrow \begin{cases} \\ -\frac{\cancel{Z_0}}{n_1}H_i\cos\theta_i + \frac{\cancel{Z_0}}{n_1}H_r\cos\theta_r = -\frac{\cancel{Z_0}}{n_2}H_t\cos\theta_t \end{cases} \Rightarrow \begin{cases} \\ -\frac{1}{n_1}(H_i\cos\theta_i - H_r\cos\theta_r) = -\frac{1}{n_2}(H_i + H_r)\cos\theta_t \end{cases}$

$-\frac{1}{n_1}(H_i\cos\theta_i - H_r\cos\theta_r) = -\frac{1}{n_2}(H_i + H_r)\cos\theta_t \Leftrightarrow -n_2 H_i\cos\theta_i + n_2 H_r\cos\theta_r = -n_1 H_i\cos\theta_t - n_1 H_r\cos\theta_t$

$\Leftrightarrow n_2 H_r\cos\theta_r + n_1 H_r\cos\theta_t = -n_1 H_i\cos\theta_t + n_2 H_i\cos\theta_i \Leftrightarrow \frac{H_r}{H_i} = \frac{-n_1\cos\theta_t + n_2\cos\theta_i}{n_2\cos\theta_r + n_1\cos\theta_t} \Rightarrow r_{TM} = \frac{-n_2\cos\theta_i + n_1\cos\theta_t}{-n_2\cos\theta_r - n_1\cos\theta_t}$

Reflexão total: $\cos\theta_i = \cos\theta_r$

$n_1\sin\theta_i = n_2\sin\theta_t \Rightarrow \sin\theta_t = \frac{n_1}{n_2}\sin\theta_i \Leftrightarrow 1 - \sin^2\theta_t = 1 - \left(\frac{n_1}{n_2}\right)^2\sin^2\theta_i \Leftrightarrow \cos^2\theta_t = 1 - \frac{\sin^2\theta_i}{\sin^2\theta_c} \Leftrightarrow \cos\theta_t = i\sqrt{\frac{\sin^2\theta_i}{\sin^2\theta_c} - 1}$

$r_{TM} = \frac{-n_2\cos\theta_i + i n_1\sqrt{\frac{\sin^2\theta_i}{\sin^2\theta_c} - 1}}{-n_2\cos\theta_i - i n_1\sqrt{\frac{\sin^2\theta_i}{\sin^2\theta_c} - 1}}$ $\qquad a = -n_2\cos\theta_i$ $\quad b = n_1\sqrt{\frac{\sin^2\theta_i}{\sin^2\theta_c} - 1}$

$r_{TM} = \frac{a + bi}{a - bi} = \frac{\cancel{\rho} e^{i\theta}}{\cancel{\rho} e^{-i\theta}} = e^{2i\theta} = e^{i\varphi_r}$

$\arg\left(\frac{a+bi}{a-bi}\right) = \arg(a+bi) - \arg(a-bi) = \operatorname{arctg}\left(\frac{b}{a}\right) - \operatorname{arctg}\left(-\frac{b}{a}\right) = 2\operatorname{arctg}\left(\frac{b}{a}\right)$

$\varphi_r = 2\operatorname{arctg}\left(\frac{b}{a}\right) \Rightarrow tg\frac{\varphi_r}{2} = \frac{b}{a}$

$\frac{b}{a} = \frac{n_1\sqrt{\frac{\sin^2\theta_i}{\sin^2\theta_c} - 1}}{-n_2\cos\theta_i} = -\frac{1}{\sin\theta_c\cos\theta_i}\sqrt{\frac{\sin^2\theta_i}{\sin^2\theta_c} - 1} = -\frac{1}{\sin\theta_c\cos\theta_i}\sqrt{\frac{\cancel{1} - \cos^2\theta_i \cancel{-1} + \cos^2\theta_c}{\sin^2\theta_c}} = -\frac{1}{\sin\theta_c}\sqrt{\frac{1}{\sin^2\theta_c}\left(-1 + \frac{\cos^2\theta_c}{\cos^2\theta_i}\right)} = -\frac{1}{\sin^2\theta_c}\sqrt{\frac{\cos^2\theta_c}{\cos^2\theta_i} - 1}$

$tg\left(\frac{\varphi_r}{2}\right) = -\frac{1}{\sin^2\theta_c}\sqrt{\frac{\cos^2\theta_c}{\cos^2\theta_i} - 1}$

$M \doteq \frac{\sin\bar{\theta}_c}{\frac{\lambda}{2d}} = \frac{\sin\bar{\theta}_c}{\frac{\lambda_0}{2n_1 d}} = \frac{0{,}3}{0{,}1} = 3 \doteq 4$

$\theta_i \to \frac{\pi}{2} - \theta$ $\qquad \theta_c \to \frac{\pi}{2} - \bar{\theta}_c$

$tg\left(\frac{\pi d}{\lambda}\sin\theta - m\frac{\pi}{2}\right) = -\frac{1}{\sin^2\left(\frac{\pi}{2} - \bar{\theta}_c\right)}\sqrt{\frac{\cos^2\left(\frac{\pi}{2} - \bar{\theta}_c\right)}{\cos^2\left(\frac{\pi}{2} - \theta\right)} - 1} = -\frac{1}{\cos^2\bar{\theta}_c}\sqrt{\frac{\sin^2\bar{\theta}_c}{\sin^2\theta} - 1}$

$g(x) = -\frac{1}{\cos^2\bar{\theta}_c}\sqrt{\frac{\sin^2\bar{\theta}_c}{\sin^2\theta} - 1}$

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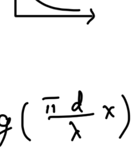

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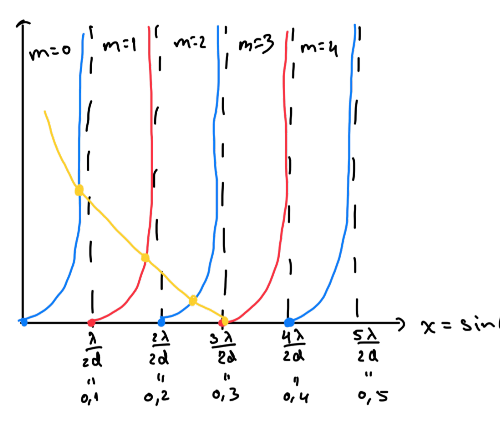


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g(x) zero: $x = \sin\bar{\theta}_c = 0{,}3$

4 Modos